MIL-C-50861A (PA)
29 March 1976
SUPERSEDING
MIL-C-50861 (MU)
16 March 1971

MILITARY SPECIFICATION

CARTRIDGE, 40MM, HEDP, M430;

BODY ASSEMBLY FOR

METAL PARTS

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the metal parts for one type of body assembly designated as Body Assembly for Cartridge, 40MM, HEDP, M430.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490	- Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specifications For

FSC: 1310

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKING DRAWINGS

9287854 - Body Assembly

INSPECTION EQUIPMENT DRAWINGS

8870372 - Ultrasonic Test Equipment

2.2 Other Publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM B 117-73 - Standard Method of Salt Spray (Fog) Testing

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103)

3. REQUIREMENTS

3.1 Material - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts - The parts shall comply with all requirements specified on Drawing (Dwg.) 9287854, all associated drawings and with all requirements specified in applicable specifications.

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3.3 Workmanship - All parts and assemblies shall be fabricated and finished in a thorough, workmanlike mannner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. All marking and stamping shall be neat and sharply defined.

3.4 First Article Inspection - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection -

4.3.1 Submission - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawings	Quantity
Body Assembly (prior to painting)	9287854	50
Bodies	9287855	20
Body Assemblies for salt spray	9287854	4
Width of coil by 20 inches from embossed stock prior to forming	9287855	2

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4.3.2 Inspections to be Performed - See MIL-A-48078 and Table 1 herein.

4.3.3 Rejection - See MIL-A-48078.

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TABLE 1 - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Body Assembly and Components				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Body (Dwg. 9287855)				
	Examination for Defects	20		3.2	4.4.2.1
	Ultrasonic Inspection	20		3.2	4.5.4
	Body Assembly (Dwg. 9287854)				
	Examination for Defects	50		3.2	4.4.2.2
	Salt Spray	#4		3.2	4.5.3
	Push Test of Rotating Band	50		3.2	4.5.5
	Embossed Stock (Dwg. 9287855)				
	Dimensional Inspection	2			4.5.2

NOTES:

Test specimen or parts.

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4.4 Quality Conformance Inspection -

4.4.1 Inspection Lot Formation - Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination - See MIL-A-48078.

a. Sampling Plans - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Body		9287855
			NEXT HIGHER ASSEMBLY
			9285854

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Presence of cracks, voids, pipes or inclusions, etc.		100%	3.2	4.5.4 and 4.5.4.1
102	Dimensional inspection of embossed stock	* (See 4.5.1)	-	3.2	4.5.2
103	Diameter between knurl		0.40%	3.2	Gage
104	Width of crest of knurl		0.40%	3.2	Gage
Minor					
201	Diameter of rotating band surface (after knurling)		0.65%	3.2	Gage
202	Width of rotating band surface		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

* Two (2) samples, width of coil by 20 inches from each coil prior to forming body.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Body Assembly, prior to Painting		9287854
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of thread		0.40%	3.2	Gage
102	Minor diameter of thread, max.		0.40%	3.2	Gage
103	Concentricity of indicated point on taper with pitch diameter of thread		0.40%	3.2	Gage
104	Diameter of rotating bands		0.40%	3.2	Gage
105	Body diameter		0.40%	3.2	Gage
106	Diameter of sealing groove		0.40%	3.2	Gage
107	Concentricity of rotating band with pitch diameter of thread		0.40%	3.2	Gage
108	Profile of rotating band		0.40%	3.2	Gage
109	Salt spray	*	*	3.2	4.5.3
110	Push test of rotating band	32	0.40%	3.2	4.5.5
Minor					
201	Total length		0.65%	3.2	Gage
202	Distance from open end to back of rotating band		0.65%	3.2	Gage
203	Length of full thread		0.65%	3.2	Gage
204	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

* Sampling and rejection shall be in accordance with TT-C-490.

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4.4.3 Testing - Testing is described in the First Article and Quality Conformance Tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Test methods and procedures.-

4.5.1 Dimensional control of stamped parts.-In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, a sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned to the contractor for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the Contracting Officer to substantiate the request.

4.5.2 Dimensional Inspection of Embossed Stock - At the start of production and for every four (4) hours production thereafter prior to forming a body, samples shall be selected from a coil of embossed stock representative of that four hours production. One sample strip shall be taken from the beginning and the end of the coil. The sample strips shall be identified as to there being from the beginning or end of the coil and also the direction of milling. If the average value for fifteen (15) measurements for center distances of

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embossment, embossed material thickness and depth of notch of the sample fails to comply with the requirements specified on the applicable drawing, corrective action shall be taken prior to further embossing.

An additional sample will be selected at the introduction of each new embossing roll to the production line. If the average value for fifteen (15) measurements for center distances of embossment, embossed material thickness and depth of notch of the sample fails to comply with the requirement specified on the applicable drawing, the embossing roll will not be used for production and corrective action taken.

The fifteen (15) measurements shall be conducted on the sample strip for each characteristic. These shall be three (3) points of measurements made across the width of the sample strip at five points along, the length of approximately every four inches. One measurement for each characteristic shall be maintained at each measuring point across the width of the material. An average value of each sample shall be completed for each characteristic from data obtained.

4.5.3 Salt Spray - This test shall be conducted in accordance with ASTM-B-117-73. This test shall be conducted for 96 hours along 1.25 inches of its outside length measured from the open end. The surface area rear of this point may be tested for 24 hours minimum. Salt spray test not required for inside surface of body assembly. The rotating band may be masked off for this test. Salt solution shall be 5 percent. This test is a destructive test. Parts so tested shall not be returned to the lot.

4.5.4 Ultrasonic Inspection - Bodies shall be inspected for the presence of cracks, voids, pipes or inclusions, etc. along the cylindrical section of the body throughout the entire wall thickness including base section, gaged externally. This test shall be performed and calibrated as specified on 8870372. If calibration shows equipment not to be performing correctly, the quantity represented prior to calibration shall be reprocessed.

4.5.4.1 Bodies shall be inspected for the presence of cracks, voids, pipes or inclusions, etc., along the base sections to a depth of .060 inches gaged externally. This test shall be performed and calibrated as specified on 8870372. If calibration shows equipment not to be performing correctly, the quantity represented prior to calibration shall be reprocessed.

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4.5.5 Push Test of Rotating Band - The body assembly shall be placed in an approved fixture and the force specified on the applicable drawing shall be applied to the rotating band. This test is a non-destructive test. Parts so tested may be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging -

5.1.1 Level C - The body assemblies shall be preserved and packaged in accordance with MIL-STD-1169.

5.2 Packing -

5.2.1 Level C - The body assemblies, preserved and packaged as specified in 5.1.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking - Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping - When components from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended Use - The components covered by this specification is intended for use in the Cartridge, 40MM, HEDP, M430.

6.2 Ordering Data - See MIL-A-48078.

6.3 Submission of Inspection Equipment Designs for Approval - See MIL-A-48078. Submit equipment designs as required to Commander, Attn: SARPA-QA-T, Picatinny Arsenal, Dover, NJ 07801.

6.4 Submission of Results of Contractor-conducted Examinations and Tests - Data shall be submitted in accordance with data item DI-R-1721 on the DD Form 1423 for the contract.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1310-A091

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